

JOINS AND SET OPERATIONS

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Create a database & use:

```
1 CREATE DATABASE IF NOT EXISTS bookflow2_db;
2 • USE bookflow_db;
```

Output

#	Time	Action
✓ 1	09:08:46	SHOW DATABASES
✓ 2	09:09:23	CREATE DATABASE IF NOT EXISTS bookflow2_db
✓ 3	09:09:40	USE bookflow_db

Create a table

```
3 • CREATE TABLE class (
4     id INT,
5     name VARCHAR(30)
6 );
7 • CREATE TABLE class_info (
8     id INT,
9     address VARCHAR(30)
10 );
```

Output

#	Time	Action
✓ 1	09:47:38	CREATE DATABASE IF NOT EXISTS bookflow3_db
✓ 2	09:47:55	USE bookflow3_db
✓ 3	09:48:21	CREATE TABLE class (id INT, name VARCHAR(30))
✓ 4	09:48:23	CREATE TABLE class_info (id INT, address VARCHAR(30))

Insert data

```
12 • INSERT INTO class VALUES
13     (1,'Likhitha'),
14     (2,'Tejaswini'),
15     (4,'Ganesh');
16 • INSERT INTO class_info VALUES
17     (1,'KERALA'),
18     (2,'ANDRA PRADESH'),
19     (3,'CHENNAI');
```

✓ 7	12:17:43	INSERT INTO class VALUES (1,'Likhitha'), (2,'Tejaswini'), (4,'Ganesh')
✓ 8	12:17:45	INSERT INTO class_info VALUES (1,'KERALA'), (2,'ANDRA PRADESH'), (3,'CHENNAI')

Show data

```
20 • SELECT * FROM class;
```

1	Likhitha
2	Tejaswini
4	Ganesh

```
21 • SELECT * FROM class_info;
```

```
22
```

1	KERALA
2	ANDRA PRADESH
3	CHENNAI

Cross join

```
22 • SELECT *  
23 FROM class  
24 CROSS JOIN class_info;  
25
```

1	Likhitha	3	CHENNAI
1	Likhitha	2	ANDRA PRADESH
1	Likhitha	1	KERALA
1	Likhitha	3	CHENNAI
1	Likhitha	2	MUMBAI
1	Likhitha	1	DELHI
2	Tejaswini	3	CHENNAI
2	Tejaswini	2	ANDRA PRADESH
2	Tejaswini	1	KERALA
2	Tejaswini	3	CHENNAI
2	Tejaswini	2	MUMBAI
2	Tejaswini	1	DELHI
4	Ganesh	3	CHENNAI
4	Ganesh	2	ANDRA PRADESH
4	Ganesh	1	KERALA
4	Ganesh	3	CHENNAI
4	Ganesh	2	MUMBAI

Inner join

```
25 • SELECT *  
26 FROM class  
27 INNER JOIN class_info  
28 ON class.id = class_info.id;
```

1	Likhitha	1	KERALA
1	Likhitha	1	DELHI
2	Tejaswini	2	ANDRA PRADESH
2	Tejaswini	2	MUMBAI
1	Likhitha	1	KERALA
1	Likhitha	1	DELHI
2	Tejaswini	2	ANDRA PRADESH
2	Tejaswini	2	MUMBAI

```

29 • SELECT class.name,
30         class_info.address
31 FROM class
32 INNER JOIN class_info
33 ON class.id=class_info.id;

```

Likhitha	KERALA
Likhitha	DELHI
Tejaswini	ANDRA PRADESH
Tejaswini	MUMBAI
Likhitha	KERALA
Likhitha	DELHI
Tejaswini	ANDRA PRADESH
Tejaswini	MUMBAI

Natural join

```

34 • SELECT *
35 FROM class
36 NATURAL JOIN class_info;

```

1	Likhitha	KERALA
1	Likhitha	DELHI
2	Tejaswini	ANDRA PRADESH
2	Tejaswini	MUMBAI
1	Likhitha	KERALA
1	Likhitha	DELHI
2	Tejaswini	ANDRA PRADESH
2	Tejaswini	MUMBAI

Left Outer join

```

37 • INSERT INTO class VALUES
38 (5,'Shiva');
39 • INSERT INTO class_info VALUES
40 (7,'NOIDA'),
41 (8,'PANIPAT');
42

```

Output			
Action Output			
#	Time	Action	
✓ 18	12:27:54	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')	
✓ 19	12:27:58	INSERT INTO class VALUES (5,'Shiva')	
✓ 20	12:28:00	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')	

Left join

1	Likhitha	1	KERALA
1	Likhitha	1	DELHI
2	Tejaswini	2	ANDRA PRADESH
2	Tejaswini	2	MUMBAI
4	Ganesh	NULL	NULL
5	Shiva	NULL	NULL

```

46 • SELECT *
47 FROM class
48 LEFT JOIN class_info
49 ON class.id=class_info.id
50 WHERE class_info.id IS NULL;

```

Right join

```

50 WHERE class_info.id IS NULL;
51 • SELECT *
52 FROM class
53 RIGHT OUTER JOIN class_info
54 ON class.id=class_info.id;

```

NULL	NULL	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

Union

```

55 • CREATE TABLE first_table(
56     id INT,
57     name VARCHAR(30)
58 );
59 • CREATE TABLE second_table(
60     id INT,
61     name VARCHAR(30)
62 );
63 • INSERT INTO first_table VALUES
64     (1,'Sita'),
65     (2,'Sai');
66 • INSERT INTO second_table VALUES
67     (2,'Sai'),
68     (3,'Ram');
69

```

```

✓ 30 12:34:31 INSERT INTO first_table VALUES (1,'Sita'), (2,'Sai')
✓ 31 12:34:33 INSERT INTO second_table VALUES (2,'Sai'), (3,'Ram')

```

Union

```

69 • SELECT * FROM first_table
70 UNION
71 SELECT * FROM second_table;
72

```

Result Grid   Filter Rows: Export

	id	name
1	1	Sita
2	2	Sai
3	3	Ram

Union All

```
72 • SELECT * FROM first_table
73 UNION ALL
74 SELECT * FROM second_table;
75
```

Result Grid | Filter Rows: Export

	id	name
▶	1	Sita
	2	Sai
	2	Sai
	3	Ram

Intersect

```
76 INTERSECT
77 ✖ SELECT * FROM second_table;
78
```

Result Grid | Filter Rows:

	id	name
▶	2	Sai

Minus

```
78 • SELECT first_table.*
79 FROM first_table
80 LEFT JOIN second_table
81 ON first_table.id=second_table.id
82 WHERE second_table.id IS NULL;
83
```

Result Grid | Filter Rows: Export

	id	name
▶	1	Sita

Address availability status:

```
82 • SELECT c.id,
83         c.name,
84         CASE
85           WHEN ci.address IS NULL
86           THEN 'Address Missing'
87           ELSE 'Address Available'
88         END AS Status
89 FROM class c
90 LEFT JOIN class_info ci
91 ON c.id=ci.id;
92
```

Result Grid | Filter Rows: Export

	id	name	Status
	1	Lakshita	Address Available
	1	Lakshita	Address Available
	2	Tejaswini	Address Available
	2	Tejaswini	Address Available
	4	Ganesh	Address Missing